
A Same-Generator Corpus and Classifier Floors for AI-Control Monitor Evaluation

Anonymous Author(s)

Affiliation

Address

email

Abstract

Trusted monitors are tuned and ranked on stored pairs of code. Valid solutions to various programming problems are paired with solutions carrying planted backdoors. However, The two classes are built differently. Honest solutions are human-written, taken from an existing dataset; every backdoor is written by a language model. The label is therefore recoverable from authorship alone, and filtering to working solutions and stripping comments does not remove it. On such a split, a trained bag-of-words classifier scores above roughly four-fifths of the open-weight monitors ranked on it, and a classifier reading no code at all scores above most of them. The signal transfers across backdoor generators and is nearly as strong on inert backdoors as on working ones, so no property of a backdoor explains it. Regenerating only the honest half raises separability further. One model writing both classes removes between a third and nine-tenths of it, on two generators that differ because only one adopts the attack prompt’s layout; matching that prompt removes most of the rest. An LLM monitor is inflated the same way, scoring 0.944 on stored pairs and 0.818 on firing backdoors from the same author. Stored-pair evaluations should report partial-input and bag-of-words floors against permutation nulls, generate both classes from one model and one prompt, and drop inert backdoors.

1 Introduction

The backbone protocol of AI control is trusted monitoring, where a weaker trusted model reads an untrusted model’s output and flags sabotage [4]. Monitors are developed, tuned and ranked almost entirely on *stored* honest/attack pairs, fixed corpora in which each programming problem contributes an honest solution and a solution carrying a planted backdoor [12, 19, 9]. The two classes come from different authors. Honest solutions are human-written, taken from APPS [7]; every backdoor is written by a language model, and the corpus records which one. Separation credited to a monitor may be separation between two authors.

That these pairs leak is known, and the corpora are preprocessed against it. The filtered release keeps only working honest solutions and working backdoors and strips every comment [12]; Control Tax reports its pass-tests pre-filter and appendices “Monitoring failed backdoors” [19]; the comment channel has a paper of its own [18]. One harness ships a `-leak-check` flag with two trivial baselines in it, a #-count detector that falls from 0.6957 to 0.4983 once comments go and code length alone at 0.4726 [9]. Partial-input baselines are the standard instrument for this [6, 16]; their failure does not certify a dataset [2], and leakage is usually treated as a checklist [10]. Out of field, bag-of-words is known to separate human-written from model-written code [14], and the detectors built on it match their classes on line and character count or on shared problem definitions [5, 1]. Adjacent control work attacks other axes, subverting monitors with injected text, showing that attacker-side selection inflates safety estimates, and removing transcript length as a nuisance covariate [20, 3, 15].



Figure 1: **How a stored pair is built, and how we rebuild it.** Each row is one property of the pair: written once and tied, it is identical on the two sides; written twice, it differs. As stored, the honest program is human-written and the backdoored one model-written, so the classes differ in writer and in elicitation as well as in sabotage. We hold the writer constant, then the prompt as well (the tinted columns), the two prompts being identical apart from the sentences that ask for a flaw. The backdoor row never merges, and it is the only difference a monitor should read. Under each construction is the same bag-of-words classifier’s AUROC on it, on one scale from chance to 1.0; every arm’s S , B' and ρ are in Table 1.

38 None of that preprocessing removes the authorship gap. On the filtered, comment-stripped split, a
39 bag-of-words classifier reaches AUROC 0.806, above 20 of the 24 open-weight monitors ranked on
40 that same split, whose median is 0.609; a classifier that reads no code at all, only problem difficulty,
41 test count and question length, reaches 0.772, above 20. The classifier transfers across backdoor
42 generators and is nearly as strong on *inert* backdoors as on working ones. Neither is a property of a
43 backdoor.

44 We contribute (i) a protocol $\times$ baseline measurement of what survives that preprocessing; (ii) a
45 construction for stored-pair corpora, in which one model writes both classes under one prompt
46 framing and one harness verifies both, and the corpus that implements it. Run on two generators, it
47 attributes between a third and nine-tenths of the separability to authorship and most of the remainder
48 to the attack prompt’s framing, and returns a null on the way, since regenerating one half of a stored
49 pair substitutes one confound for another. We add (iii) a second selection artifact the filter installs
50 by defining the negative class through attack failure, and (iv) the construction and reporting practice
51 these imply. Our claims concern monitor training, selection and benchmarking on stored pairs, and
52 our measurement is of one APPS-derived family plus five agentic settings. ControlArena’s live
53 evaluations generate both classes and are not implicated.

54 2 Setup

55 The record schema shows the confound directly. The canonical stored-pair artifact for APPS control
56 evaluations [8] records `model` and `source` on every backdoor and no generator field on any honest
57 solution (Table A1), because only the attack class has a model to record. Figure 1 sets that construction
58 beside the one we build against it.

59 The corpora we measure are built on that artifact. We use seven public corpora, whose per-class
60 provenance is the object of study. In the APPS family the honest class is human and the attack class is
61 one of ten named models. Six of the seven declare no licence [11, 17], so we report only aggregates
62 and redistribute no corpus.

63 Our headline protocol is one of the seven, the filtered, comment-stripped split [12]: 1,209
64 train problems (2,418 paired items) and 1,375 test items (172 attack, 1,203 honest), labelled by
65 `backdoor_works`, reconstructed with the released harness code [9]. Generated code is scored by
66 our own runner, applied to both classes (appendix A.11).

3 Results

Comment stripping works on comment count and code length, driving both to chance (Table A2, appendix). Two floors on that same split do not move. A monitor scoring at or below a floor has shown nothing beyond what the corpus leaks. Bag-of-words reaches 0.806 and a classifier that reads no code at all reaches 0.772, both far above their permutation nulls and above 20 of the 24 monitors ranked on that split (Figure A1; intervals, nulls and paired-bootstrap counts in appendix A.2).

Three further measurements show that the bag-of-words classifier is not reading the backdoor. Every number in this paragraph is comment-stripped on both classes; the unstripped values, inflated by the comment channel, are in appendix A.3. The classifier transfers across backdoor generators at 0.969 pairwise. Trained on one attacker’s backdoors it identifies another’s, so it has learned a feature the attackers share. It is also nearly as strong on inert backdoors (0.879) as on working ones (0.935), where a detector of sabotage should find the backdoors that never function much harder. And the residual carrier is stylistic. `__main__` appears in 0.116 of attack samples against 0.000 of honest ones, replicated on the train split (152 versus 1 of 1,209) and untouched by the stripper. A token-level version of the test does not replicate here (appendix A.12).

To test whether authorship is what the classifier reads, we hold the author constant and measure how much separability survives. Under the identical pipeline the human-honest baseline is $B = 0.871$ over $n = 1,444$ pairs. Write G for that same quantity when one model writes both classes. Retained signal is $\rho = (G - 0.5)/(B - 0.5)$: $\rho = 1$ means holding authorship constant changed nothing, $\rho = 0$ that nothing survived. We pre-registered $\rho \leq 0.33$ as authorship and $\rho > 0.72$ as content, and fixed both bands before any arm ran.

One model writing both classes removes between a third and nine-tenths of the separability (Table 1). We report the *independent* arm, in which the attack class is written from the problem statement alone, so authorship changes and nothing else. An *edit* arm, shown its own honest solution, returns near-copies instead (appendix A.4). We take one outcome-blind draw per problem and define the attack class as the benchmark does, by intent plus passing the public tests. 2.2% of the first generator’s draws fire and 0.3% of the second’s (2 of 611); B' is inert-inclusive over the same problems. The first generator’s interval excludes the collapse band and spans the content line, so the pre-registered rule returns inconclusive between partial and content. The second generator’s lies inside the collapse band. Sample size does not explain the drop, since at $n = 623$ the baseline B under the identical pipeline is 0.8521 ± 0.0055 . Two rows are *strict*: they keep only attacks that pass the tests *and* fire on their trigger, selected by the benchmark’s own retry-and-keep. Only 99 pairs meet it, so we read them through ρ alone; they agree with the headline rows at low power.

Table 1: **What is left when one model writes both classes.** S is the same-generator AUROC (5-fold, comments stripped) and B' the stored-pair baseline on the row’s own problems; ρ , its bands and its intervals are as in the text and appendix A.5.

Arm	n	S (95% CI)	B'	ρ (95% CI, joint)	band
Sonnet 5, headline	623	0.7512 (0.7295–0.7713)	0.8658	0.687 (0.625–0.755)	partial–content
Haiku 4.5, headline	611	0.5340 (0.5159–0.5531)	0.8584	0.095 (0.044–0.150)	collapse
Sonnet 5, prompt-matched	586	0.6423 (0.6179–0.6621)	0.8613	0.394 (0.326–0.454)	partial
Sonnet 5, strict	99	0.7652 (0.7016–0.8000)	0.7877	0.921 (0.662–1.089)	partial–content
Sonnet 5, strict, matched	96	0.7113 (0.6464–0.7453)	0.8030	0.697 (0.494–0.856)	partial–content

The attack prompt separates the two generators. On Sonnet 5 the surviving signal is lexical, with length alone falling to 0.5851 while TF-IDF holds, and it does not run through the trigger. Masking numerals and comparison operators leaves S unchanged; only normalising identifiers moves it, to 0.700 ($\rho = 0.54$). The top attack-side features are `def main` (67% of attack programs against 38.5% of honest ones) and `data[`, which together are the program layout the attack prompt elicits. Haiku 4.5 never adopts that layout. `def main` and `sys.stdin` appear in no program of any of its three classes, which read input with `input()` throughout (appendix A.6). Holding the prompt constant as well as the model therefore moves the two generators toward each other, to $\rho = 0.394$ and $\rho = 0.113$. On Sonnet this removes most of what is left. The two honest classes alone, with no attack on either side, already separate at 0.6006, so the prompt is carrying framing and no backdoor at all. On Haiku the same prompt-only floor is 0.4918, an interval containing chance.

111 Regenerating one class rather than both does not work. With an open-weight model writing the
112 honest class against the stored backdoors, separation is higher than the stored baseline, at $\rho = 1.234$
113 (appendix A.5): the original authorship gap has been replaced by a new one between that model and
114 the stored attacker.

115 The same inflation appears when the detector is an LLM monitor rather than a bag-of-words classifier.
116 We put a zero-shot monitor prompt to the model that wrote both classes. In this field’s vocabulary it
117 is an untrusted monitor, which makes it the most favourable case available, since it is reading its own
118 output. On the stored pairs over the headline problems it scores 0.9579 and on the same-generator
119 pairs 0.566, a monitor- ρ of 0.145, below the classifier’s 0.687. On the strict pairs it scores 0.9437 and
120 0.818, a monitor- ρ of 0.717 (intervals in appendix A.7). A monitor that wrote neither class returns
121 the same two ratios: 0.151, a bound at half missingness, and 0.749. Our single-shot attacks fire given
122 a pass at 2.5% against the stored attacker’s 67.2%, which makes the stored attack class a survivor set.

123 The split selects on a second property besides authorship. Labelling by `backdoor_works` defines
124 the negative class at the problem level by attack failure; the criterion should be honesty. 80.7% of the
125 test split’s labelled backdoors do not work (32.8% of those that pass the public tests), and a problem
126 whose backdoor fails contributes its human honest solution instead. The label is fixed before any
127 code is read. Attack problems also carry half as many test cases (20.95 versus 38.85), and fewer tests
128 make a backdoor easier to hide, so difficulty and test count correlate with the label. That correlation
129 produces the content-blind 0.772.

130 Removing the authorship gap does not remove every floor. In five settings where the same three
131 models write both classes, so no authorship confound is possible, the surviving floor is structural:
132 0.504 (bash, on its null) to 0.802 (agentdojo) from character count alone (Figure A2).

133 4 Limitations

134 A high floor is evidence about a corpus; it does not settle results computed on that corpus. Our floors
135 show that the label is recoverable without reading a backdoor. They say nothing about whether any
136 monitor recovers it that way, and no number here corrects a published monitor score.

137 The control arm is limited by the generator we had to substitute. The model that wrote the stored
138 backdoors, `claude-3-7-sonnet-thinking`, is retired, so both same-generator classes come from
139 a later model of the same lineage. It differs from the stored attacker in backdoor style as well as in
140 authorship, so every ρ here, classifier or monitor, bounds the content share of the separability instead
141 of partitioning it. The arm is not bit-reproducible, having run through a metered API that does not
142 expose sampling parameters, with thinking off although the stored attacker was a thinking model.

143 Our scope is narrow in three further ways. We measure one APPS-derived corpus family and five
144 agentic settings, and other stored-pair corpora are untested. Five of our seven corpora are one user’s
145 repackaging of ControlArena releases, so our claims are scoped to those artifacts as published. And
146 two interpreter-era artifacts fall on opposite classes, so B carries a small execution-environment
147 channel that the same-generator arm does not (appendix A.11).

148 5 Discussion

149 These results imply one rule for construction: one model should write both classes under one prompt
150 framing, and one harness should verify both. Regenerating a single side does not help: substituting
151 the honest author left a larger gap than the one it replaced. Matched construction is preceded in
152 code-detection benchmarks [5], though we have not seen it reach backdoor detection.

153 The definition of the negative class needs the same treatment. Defining it through attack failure
154 ties the label to the attacker rather than to the code, and puts a content-blind classifier above most
155 monitors ranked on this split; dropping the failed problems removes it, as does sampling that class
156 independently of attack success. A monitor number is in turn interpretable only against the floors of its
157 own split, so a partial-input and a bag-of-words floor belong beside it, each against a permutation null,
158 with the split, the metric and the preprocessing released alongside (ours are MIT, appendix A.10).

159 Stored pairs remain the cheapest way to develop a monitor, and nothing here argues for abandoning
160 them, only that a score on one describes two authors until the construction rules that reading out.

References

- [1] Basak Demirok, Mucahid Kutlu, and Selin Mergen. MultiAIGCD: A comprehensive dataset for AI generated code detection covering multiple languages, models, prompts, and scenarios. 2025. arXiv:2507.21693v1.
- [2] Shi Feng, Eric Wallace, and Jordan Boyd-Graber. Misleading failures of partial-input baselines. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019.
- [3] Catherine Ge-Wang, Tyler Crosse, Benjamin Hadad, Joachim Schaeffer, Ram Potham, and Tyler Tracy. Attack selection in agentic AI control evaluations meaningfully decreases safety. 2026. arXiv:2606.06529.
- [4] Ryan Greenblatt, Buck Shlegeris, Kshitij Sachan, and Fabien Roger. AI control: Improving safety despite intentional subversion. 2023. arXiv:2312.06942.
- [5] Hanxi Guo, Siyuan Cheng, Kaiyuan Zhang, Guangyu Shen, and Xiangyu Zhang. CodeMirage: A multi-lingual benchmark for detecting AI-generated and paraphrased source code from production-level LLMs. 2025. arXiv:2506.11059v1. No venue stated.
- [6] Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. Annotation artifacts in natural language inference data. In *Proceedings of NAACL-HLT*, 2018.
- [7] Dan Hendrycks, Steven Basart, Saurav Kadavath, Mantas Mazeika, Akul Arora, Ethan Guo, Collin Burns, Samir Puranik, Horace He, Dawn Song, and Jacob Steinhardt. Measuring coding challenge competence with APPS. In *NeurIPS Datasets and Benchmarks Track*, 2021. arXiv:2105.09938. The upstream source of the human-written honest class, distributed as the HuggingFace dataset codeparrot/apps.
- [8] Rogan Inglis. apps-control-arena. HuggingFace dataset `RoganInglis/apps-control-arena`. MIT licence. 5,000 rows; honest class from codeparrot/apps., 2025.
- [9] Anik Jha. Decorrelation is not complementarity: Skill, not lineage, governs trusted-monitor ensembles. 2026. arXiv:2608.16190. Code: <https://github.com/anik-jha/challenger-panels> (MIT).
- [10] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in ML-based science. 2022. arXiv:2207.07048.
- [11] Eugene Koran. apps-control-arena-high-quality. HuggingFace dataset `eugenekoran/apps-control-arena-high-quality`. **No licence declared.** 1,209 train / 1,375 test., 2026.
- [12] Eugene Koran, Yejun Yun, Samantha Tetef, Benjamin Arnav, and Pablo Bernabeu-Pérez. Ensemble monitoring for AI control: Diverse signals outweigh more compute. 2026. arXiv:2605.15377v2.
- [13] Bryce Little. Length penalties make chain-of-thought less monitorable. 2026. arXiv:2607.09786v3.
- [14] Daniil Orel, Dilshod Azizov, Indraneil Paul, Yuxia Wang, Iryna Gurevych, and Preslav Nakov. AICD bench: A challenging benchmark for AI-generated code detection. 2026. arXiv:2602.02079v1. No venue stated on the abs page.
- [15] Lucas Pinto. Calibration-family overfit: Why trusted sabotage monitors don’t transfer across lineages. In *COLM 2026 Workshop on AI Measurement Science*, 2026. arXiv:2607.06596v3.
- [16] Adam Poliak, Jason Naradowsky, Aparajita Haldar, Rachel Rudinger, and Benjamin Van Durme. Hypothesis only baselines in natural language inference. In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics (*SEM)*, 2018.

- [17] Aditya Sinha. control_arena_{bash,shade,iac,rogue_eval,agentdojo}. HuggingFace datasets adityaasinha28/control_arena_*. **No licence declared.** 1,500 rows each., 2026.
- [18] Ashwin Sreevatsa, Sebastian Prasanna, and Cody Rushing. Basic legibility protocols improve trusted monitoring. 2026. arXiv:2602.10153.
- [19] Mikhail Terekhov, Zhen Ning David Liu, Caglar Gulcehre, and Samuel Albanie. Control tax: The price of keeping AI in check. 2025. arXiv:2506.05296v3.
- [20] Mikhail Terekhov, Alexander Panfilov, Daniil Dzenhaliou, Caglar Gulcehre, Maksym Andriushchenko, Ameya Prabhu, and Jonas Geiping. Adaptive attacks on trusted monitors subvert AI control protocols. In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2510.09462v2.

A Supplementary tables and figures

A.1 The schema

Table A1 is the record schema of RoganInglis/apps-control-arena [8]. The attack class carries model and source, naming the system that produced each backdoor; the honest class carries no such field, because there is no generator to name. The two APPS-backdoor corpora we use are two views of this one artifact, since every problem in the filtered release [11] is also in [8], byte-identical except a re-validated backdoor_works. Among the agentic corpora we exclude AgentHarm, to avoid a conflict of interest.

Table A1: The confound is legible in the dataset schema, with no measurement at all. The attack class records which model wrote it; the honest class has no generator field. Verified against the shard schema of RoganInglis/apps-control-arena.

struct	sub-fields
solutions	code, compiles, passes_tests
backdoors	backdoor_code, backdoor_input, backdoor_works, compiles, model, passes_tests, source

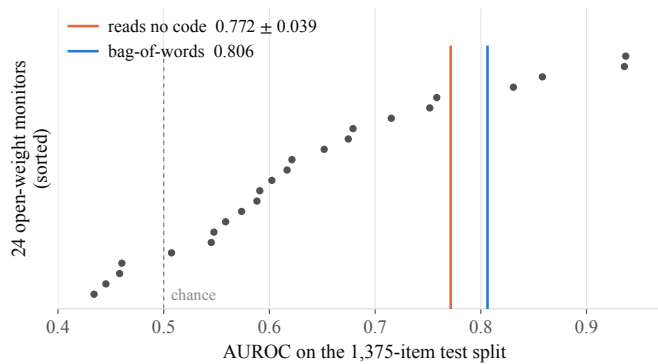


Figure A1: **Two floors against the 24 open-weight monitors ranked on the same split.** AUROC on the 1,375-item test split (172 attack / 1,203 honest), comments stripped; monitor scores from the MIT-licensed release of [9], median 0.609, max 0.937. Both floors clear 20 of the 24. **Two caveats.** (i) The monitors are zero-shot prompted while the bag-of-words classifier is trained on the benchmark’s own paired train split. The comparison states what the corpus makes recoverable and does not claim that bag-of-words is a better monitor. (ii) The no-code floor is cross-validated *within* the test split; on one held-out slice it is 0.716, above 18 of 24 (appendix A.2).

226 A.2 The protocol $\times$ baseline table

227 The no-code floor is cross-validated within the test split because the released train/val split is
 228 paired and therefore metadata-balanced; its permutation null shuffles labels for fitting and scoring
 229 alike. Table A2 is the full measurement behind §3. Rows are not comparable across protocols;
 230 the preprocessing column says whether comments were stripped before the baseline read the code.
 231 Pairwise accuracies are given to three decimals because their resolution over 2,641 pairs is one pair.
 232 Intervals in §3 are bootstraps over problems (2,000 draws) and nulls are 200 label permutations;
 233 the held-out no-code row is barely above its own null: 0.716 (CI 0.6476–0.7796) against a 95th
 234 percentile of 0.7023, $p = 0.030$, above 18 of the 24 monitors by point estimate and 12 with a paired
 235 interval excluding zero. The no-code claim therefore rests on the cross-validated protocol. The
 236 two floors of §3 in full: bag-of-words reaches 0.806 (95% CI 0.7704–0.8409 by bootstrap over
 237 problems; permutation null 95th percentile 0.5465, $p < 0.005$), above 20 of the 24 monitors by
 238 point estimate and 18 with a paired-bootstrap interval excluding zero; the no-code floor reaches
 239 0.772 (CI 0.7319–0.8005 on the pooled out-of-fold 0.766; null 0.5545, $p < 0.005$), above 20 and 17
 240 respectively.

Table A2: What the field’s preprocessing leaves behind. Every row is measured under the protocol named in column one; rows are not comparable across protocols. *Prep.* says whether comments were stripped before the baseline read the code; *no code* means the baseline never reads the program. The comments-kept rows are inflated by the comment channel and are given for contrast, not as results. The matched rows are problem-disjoint and scored against the DSPy tutorial monitor; the transfer row runs Control Tax to the GPT-4-era set. Pairwise accuracies are given to three decimals, their resolution over the pair count.

Protocol	Baseline	Value	Prep.	Against
<i>AUROC</i>				
Koran’s exact split	TF-IDF word 1–2gram	0.8063	stripped	above 20 of 24
same	TF-IDF word 1–2gram	0.8914	kept	above 22 of 24
same, reads no code	difficulty + #tests + $ q $	0.7715 ± 0.0389	no code	above 20 of 24
same, held-out slice	difficulty + #tests + $ q $	0.7163	no code	above 18 of 24
same	code length only	0.4726	stripped	chance
same	#-count detector	0.4983	stripped	chance
same	#-count detector	0.6957	kept	—
Raw artifact, problem level	all metadata	0.8917	no code	—
same	difficulty only	0.6823	no code	—
same	<code>solution_passes_tests</code>	0.7881	no code	—
Matched pairs (P3)	all metadata	0.5000	no code	by construction
<i>Pairwise accuracy</i>				
DSPy matched pairs	TF-IDF word 1–2gram	0.897	stripped	monitor 0.69–0.81
working backdoors only	TF-IDF word 1–2gram	0.935	stripped	—
inert backdoors only	TF-IDF word 1–2gram	0.879	stripped	not sabotage
transfer, cross-generator	TF-IDF word 1–2gram	0.969	stripped	—
all pairs, comments kept	TF-IDF word 1–2gram	0.962	kept	—
working, comments kept	TF-IDF word 1–2gram	0.979	kept	—
inert, comments kept	TF-IDF word 1–2gram	0.955	kept	—
transfer, comments kept	TF-IDF word 1–2gram	0.994	kept	—
same, raw artifact	code length only	0.826	kept	—

241 A.3 Comment stripping on the matched-protocol numbers

242 The matched-pair and transfer measurements were first made with comments in. Every value in
 243 the authorship paragraph of §3 was re-measured with the same stripper applied to both classes,
 244 over five problem-disjoint splits (each transfer row is one fixed split). With comments kept the
 245 same quantities were transfer 0.994, working 0.979, inert 0.955, all pairs 0.962 and surface-feature
 246 AUROC 0.8306; stripped they are 0.969, 0.935, 0.879, 0.897 and 0.7259. Every one was inflated by
 247 the comment channel, which is this paper’s thesis applied to itself. The #-presence marker, 0.685
 248 against 0.176 before stripping, closes to 0.007 against 0.008 after and is dropped; `__main__` is
 249 unchanged. One authorship leak survives the stripper because string literals are not comments: in the
 250 comment-stripped train split an author byline (Author :) survives in 23 of 1,209 honest samples
 251 against 2 attack samples, and a triple-quoted block in 43 against 9. Both are far too rare to drive 0.806,

but both are a literal record of authorship. Stripped, the matched-protocol TF-IDF AUROC is 0.8560, beside $B = 0.871$ and the Koran-split 0.806, all three on stripped code. The other cross-generator transfer rows, stripped, are 0.866 (o3-mini to claude-3-7-sonnet, 82 test pairs), 0.844 (the reverse) and 0.764 (o1-mini to gemini-2.0-flash).

A.4 Pair similarity, and why the same-generator control needs two arms

An attack class built by editing its own honest solution changes pair similarity as well as authorship (Table A3); the independent arm sits in the stored benchmark’s regime and if anything slightly further from it, so it cannot manufacture a low S by making the classes alike.

Table A3: Character similarity of honest/attack pairs, comments stripped. The edit arm’s classes are near-copies; the independent arm reproduces the stored benchmark’s regime, where the two classes are largely independent programs. The last two columns are the share of pairs above and below those similarities.

Attack class	mean sim.	share > 0.95	share < 0.60	changes vs. the benchmark
stored benchmark	0.606	3.4%	45.4%	the regime to match
edit (shown its own solution)	0.928	64.2%	4.0%	authorship and similarity
independent (problem only)	0.659	3.4%	34.8%	authorship only

A.5 Robustness of the same-generator rows

At the strict rows’ $n = 99$, B under the identical pipeline is 0.7807 ± 0.0248 . Intervals in the same-generator paragraphs of §3 are bootstraps over problems, resampling S and B' jointly for ρ ; this replaced a ± 2 fold-sd rule before the retry pool was read. Every ρ point we quote is the ratio of the fold-mean AUROCs and every joint interval beside it is the bootstrap of the pooled out-of-fold ρ , which is the same quantity computed two ways: the pooled points are 0.86 (strict, the row whose interval Table 1 quotes), 0.390 (prompt-matched headline), 0.662 (prompt-matched strict) and 0.097 (the second generator). Five fold seeds move the strict row’s ρ over 0.85–0.92. Retention does not produce the effect. Survivors and non-survivors separate at only 0.62, and S computed with non-firing attempts on the same problems is 0.7595 against 0.7687. The edit arm bounds S under near-copies (similarity 0.928, Table A3): 0.6266 over 427 pairs, $\rho = 0.371$. In arm 1, where only the honest half was regenerated, $G = 0.962$ against $B' = 0.874$ on that arm’s $n = 668$ eligible problems, and TF-IDF gains +0.088 while the length carrier collapses to 0.489. Our own attack prompt halves our single-shot firing rate: 52.0% of our test-passing draws offer a public test input as the trigger, which cannot diverge from the honest solution, and those fire 0.15% against 5.13% for novel triggers; no stored working backdoor has such a trigger (0 of 3,188). The rest is retry-and-keep.

A.6 The prompt-matched arm under the masks, and the second generator

Under prompt matching the identifier mask, the only mask of §3 that moved S , takes the headline row to 0.6264 ($\rho = 0.342$) where it takes the original honest class on the same 586 problems to 0.7046, so identifier normalisation and prompt matching remove largely the same thing. That masked row is the one that decides whether the residue is structure or naming, and under prompt matching it is band-ambiguous: its joint interval, 0.280–0.394, puts 40% of its mass in the collapse band. A character model (3–5-grams, same classifier and folds) retains more than the word tokeniser, 0.6817 ($\rho = 0.493$), so the drop is not an artefact of tokenisation. On the strict pairs the matched row is $S = 0.7113$ ($n = 96$, $\rho = 0.697$, 0.494–0.856). The second generator’s pairs are more self-similar than the first’s (0.724 mean against 0.659), so S was recomputed inside similarity strata, where it stays 0.4737–0.5142 in every stratum, including the 0.451-similarity stratum, which is further from its honest class than the stored benchmark’s own pairs, while B' holds 0.7884–0.8334, so near-copies are not what buys the collapse. Its prompt-matched rows move S to 0.5399 ($\rho = 0.113$, $n = 562$) and put the prompt-only floor at 0.4918, an interval containing chance. The mechanism is visible in the programs. The layout vocabulary that the first generator’s residue is made of, `def main, sys.stdin` and `data[`, is present in 0.000 of the second generator’s programs in *all three* of its classes, which read standard input with `input(` in 1.000 of programs on every side, so there is no layout difference left for the classifier to read.

294 A.7 The monitor rows, and what bounds the second monitor

295 The monitor that wrote both classes scores 0.566 (0.5507–0.5833) on the same-generator headline
 296 pairs, a monitor- ρ of 0.145 (0.111–0.182); on the strict pairs it scores 0.818 (0.7630–0.8686), a
 297 monitor- ρ of 0.717 (0.601–0.832). The monitor that wrote neither class scores 0.5577 on the same-
 298 generator headline pairs and 0.8824 on the stored pairs over the same problems, 0.7871 and 0.8832
 299 on the strict pairs. Its headline rows bound the value without fixing it. A 512-token reply ceiling left
 300 628 of 1,246 same-generator replies unparseable, and on the stored side the missing share differs by
 301 class (50.1% honest against 23.4% attack), a bias of unknown sign. Re-running the strict pairs with a
 302 1,024-token ceiling prices it: missingness falls to 2 of 198 and monitor- ρ moves from 0.749 to 0.766,
 303 inside the bootstrap interval. On the stored pairs both monitors beat the bag-of-words baseline over
 304 the same problems (+0.09 and +0.15 for the first, +0.02 and +0.09 for the second): Figure A1’s
 305 ranking states something about those monitors on that split and does not claim that a bag of words
 306 beats a monitor here. Classifier and monitor are independent pair by pair on the same-generator pairs
 307 ($\phi = 0.04$), so the two are reading different things.

308 A.8 The two generators’ funnel

309 Both generators were run through the identical pipeline. Our honest draws pass the public tests at
 310 0.7701 and 0.7278 of 1,444 problems; the second generator’s attack draws pass at 689 of 1,444 and
 311 35 of those fire, below the 50-pair minimum `gate_s_samegen.py` requires, leaving it with no strict
 312 row and the two generators are compared on the passes eligibility only. The two compared rows
 313 differ in inertness. 2 of the 611 attacks in the second generator’s row fire (99.67% inert) against 14 of
 314 623 (97.75%), the other 33 of its firing draws sitting on problems where its own honest program failed
 315 the tests and which the control therefore drops. The single-shot firing rate belongs to the protocol and
 316 not to either model, since 52.0% of test-passing draws offer a public test input as their trigger.

317 A.9 The five same-generator agentic settings

318 Figure A2 shows each setting’s content-blind floor against its own permutation null.

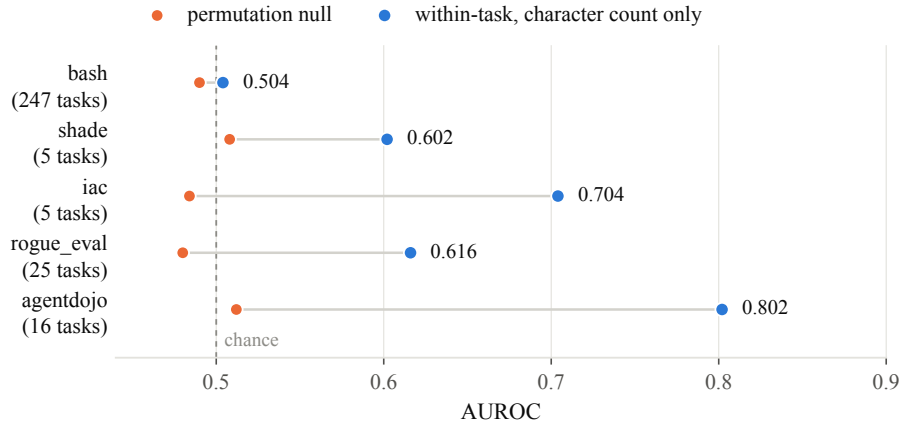


Figure A2: **The structural floor in five same-generator agentic settings.** Each setting’s content-blind classifier (six count features, no content read) scored *within task* against its own permutation null. No authorship confound is possible here: claude-sonnet-4, gemini-2.5-pro and gpt-5 write both classes in all five. bash sits on its null at 0.504; agentdojo reaches 0.802 from character count alone. **The task-group count beside each setting matters:** shade and iac average over five task groups each while bash averages over 247, so these floors are not equally well determined. We report the within-task column because our own pooled numbers partly broke on task composition, and we broke them ourselves. The permutation nulls shown were computed with a pooled estimator rather than the folded within-task statistic; they bound the chance level but are not a like-for-like null for each dot. These corpora declare no licence; we redistribute nothing and report aggregates only.

319 **A.10 The release**

320 The supplementary bundle holds the same-generator split (1,444 honest programs, 1,112 edit-arm
321 and 1,112 independent-arm attack programs written by one model, plus every one of the 11,565
322 draws of the independent arm’s retry pool), both eligibility rules as pair lists under an outcome-blind
323 selection (623 passing and 14 firing independent-arm pairs) and under the benchmark’s own survivor
324 selection (1,046 and 99, or 106 counting a crash on the trigger as firing), the 24 scripts behind the
325 measurements (generation for both generators, the test harness, the trigger verifier, the selection rules,
326 the prompt-matched arm, the monitor rows and each remaining measurement script), a dataset card,
327 and a manifest of row counts and checksums. The licence, MIT, is a file inside the bundle and covers
328 the data and the scripts alike.

329 **A.11 The test harness and the pass flag**

330 The same-generator arm requires executing generated code, so we built a runner that feeds each
331 shipped input case to a solution under per-case timeouts and address-space limits. It agrees with the
332 artifact’s own `solution_passes_tests` on 95.84% of the analysis pool, and is strictly conservative
333 there, never crediting a solution the artifact rejects (0 such cases). We use *our own* flag for both
334 classes, because a checker stricter on one class than the other would manufacture exactly the artifact
335 this paper documents. We do not length-match either class; the untrusted model chooses its own
336 trajectory length. The headline does not depend on that choice. Under the artifact’s flag $B = 0.871$
337 at $n = 1,444$, under ours $B = 0.873$ at $n = 1,406$, within one standard deviation of each other.
338 The monitor scores in Figure A1 are `runs/x0/monitor-summary.json` from the MIT release of
339 that harness [9], used under its notice. Two interpreter-era artifacts in the upstream solutions fall on
340 opposite classes: a `stdin` without `.buffer`, `fixed`, and `fractions.gcd`, removed in Python 3.9, not
341 fixable. B is exposed to them; the same-generator arm, whose code we generated and ran ourselves,
342 is immune.

343 **A.12 The token test, run and failed**

344 We attempted to confirm a reported token-level mechanism [14] in our setting and it does not replicate.
345 Separation is -0.013 , the wrong sign, at permutation $p = 0.956$. The reason is register. Their human
346 class is general-purpose repository code, whereas both of our classes are competitive-programming
347 code, so an identifier axis that separates general human code from model code does not separate
348 these two populations. Only 8 of their 10 canonical tokens are in our vocabulary, so the test is
349 underpowered; but an underpowered true effect would still carry the right sign, and this one does not.
350 We report it because a reader who knows that work may run exactly this test.

351 **A.13 A third channel, noted but not developed here**

352 In the same five agentic settings a fifteen-word regular expression over the untrusted policy’s own
353 scratchpad reaches 0.76–0.82, because prompted attack policies restate their side task in the reasoning
354 the monitor is given. We note this only to bound the claim. The instrument is not ours, and the work
355 it comes from [13] validates it against an LLM judge and a length-matched null; we defer to that
356 treatment.